

$$\begin{aligned}
 1. \quad W_t &= (1-B)(1-B^{12})X_t \\
 &= (1-B)(X_t - X_{t-12}) \\
 &= (1-B)[a + bt + s_t + Y_t \\
 &\quad - (a + b(t-12) + s_{t-12} + Y_{t-12})] \\
 &= (1-B)[12b + Y_t - Y_{t-12}] \\
 &= 12b + Y_t - Y_{t-12} \\
 &\quad - (12b + Y_{t-1} - Y_{t-13}) \\
 &= Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}
 \end{aligned}$$

(a) Y_t is a white noise process.

$$\begin{aligned}
 \text{Var}(W_t) &= \text{Var}(Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}) \\
 &= \text{Var}(Y_t) + \text{Var}(Y_{t-1}) \\
 &\quad + \text{Var}(Y_{t-12}) + \text{Var}(Y_{t-13})
 \end{aligned}$$

「because $\{Y_t\}$ is a white noise process」

$$\gamma_W(0) = 4\sigma_Y^2$$

Consider

$$\gamma_W(h) = \text{Cov}(W_t, W_{t+h}) \quad h > 0.$$

$$\begin{aligned}
 &= \mathbb{E}[(Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}) \\
 &\quad \times (Y_{t+h} - Y_{t+h-1} - Y_{t+h-12} + Y_{t+h-13})]
 \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E}[Y_t Y_{t+h}] - \mathbb{E}[Y_t Y_{t+h-1}] - \mathbb{E}[Y_t Y_{t+h-12}] \\
&\quad + \mathbb{E}[Y_t Y_{t+h-13}] \\
&\quad - \mathbb{E}[Y_{t-1} Y_{t+h}] + \mathbb{E}[Y_{t-1} Y_{t+h-1}] \\
&\quad + \mathbb{E}[Y_{t-1} Y_{t+h-12}] - \mathbb{E}[Y_{t-1} Y_{t+h-13}] \\
&\quad - \mathbb{E}[Y_{t-12} Y_{t+h}] + \mathbb{E}[Y_{t-12} Y_{t+h-1}] \\
&\quad + \mathbb{E}[Y_{t-12} Y_{t+h-12}] - \mathbb{E}[Y_{t-12} Y_{t+h-13}] \\
&\quad + \mathbb{E}[Y_{t-13} Y_{t+h}] - \mathbb{E}[Y_{t-13} Y_{t+h-1}] \\
&\quad - \mathbb{E}[Y_{t-13} Y_{t+h-12}] + \mathbb{E}[Y_{t-13} Y_{t+h-13}]
\end{aligned}$$

$$\begin{aligned}
h=1: \quad \gamma_w(h) &= -\mathbb{E}(Y_t^2) - \mathbb{E}(Y_{t-12}^2) \\
&= -2\sigma_z^2
\end{aligned}$$

$$h=11: \quad \gamma_w(h) = \mathbb{E}(Y_{t-1}^2) = \sigma_z^2$$

$$\begin{aligned}
h=12: \quad \gamma_w(h) &= -\mathbb{E}(Y_{t-1}^2) - \mathbb{E}(Y_t^2) \\
&= -2\sigma_z^2
\end{aligned}$$

$$\begin{aligned}
h=13: \quad \gamma_w(h) &= \mathbb{E}(Z_t^2) \\
&= \sigma_z^2
\end{aligned}$$

$\gamma_w(h)$:

$$\gamma_w(h) = \begin{cases} 4\sigma_z^2 & h=0 \\ -2\sigma_z^2 & |h|=1 \\ \sigma_z^2 & |h|=11 \\ -2\sigma_z^2 & |h|=12 \\ \sigma_z^2 & |h|=13 \\ 0 & \text{otherwise.} \end{cases}$$

$$\rho_w(h) = \frac{\gamma_w(h)}{\gamma_w(0)}.$$

2(a) Consider $k=1, p=1$:

$$X_t = \phi_1 X_{t-1} + Z_t$$

$$\Rightarrow (1 - \phi_1 B) X_t = Z_t$$

Set

$$(1 - \phi_1 B) \hat{X}_{n+1} = [Z_{n+1}]$$

$$[Z_k] = \begin{cases} 0 & k > n \\ Z_k & k \leq n \end{cases}$$

「FUTURE」
「PAST/PRESENT」

「Seek forecasted value of X_t at $p+n$ where $p=1$ (obs. series to n)」

$$\downarrow \hat{X}_{n+1}(n)$$

$$(1 - \phi_1 B) \hat{X}_{n+1} = 0$$

$$\Rightarrow \hat{X}_{n+1} - \phi_1 \hat{X}_n = 0$$

$$\Rightarrow \hat{X}_{n+1} = \phi_1 X_n.$$

$$p=2, k=1$$

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

$$\Rightarrow (1 - \phi_1 B - \phi_2 B^2) X_t = Z_t$$

「Seek $\hat{X}_{n+1}(n)$ 」

$$(1 - \phi_1 B - \phi_2 B^2) \hat{X}_{n+1} = [Z_{n+1}]$$

$$[Z_k] = \begin{cases} 0 & k > n \\ Z_k & k \leq n. \end{cases}$$

$$\therefore (1 - \phi_1 B - \phi_2 B^2) \hat{X}_{n+1} = 0$$

$$\Rightarrow \hat{X}_{n+1} - \phi_1 X_n - \phi_2 X_{n-1} = 0$$

$$\Rightarrow \hat{X}_{n+1} = \phi_1 X_n + \phi_2 X_{n-1}$$

(b) From lectures, the mean squared prediction error is

$$\sigma_k^2 = \sigma_z^2 \sum_{j=0}^{k-1} b_j^2$$

$\{b_j\}$ terms are from the $MA(\infty)$ representation of $\{X_t\}$

i.e. $X_t = \sum_{j=0}^{\infty} b_j Z_{t-j}$

Note $X_t = (1 - \phi_1 B)^{-1} Z_t$

$$= \sum_{j=0}^{\infty} \phi_1^j Z_{t-j} \quad \text{Taylor series exp.}$$

$$\therefore \sigma_k^2 = \sigma_z^2 \sum_{j=0}^{k-1} (\phi_1^j)^2$$

$$= \sigma_z^2 \sum_{j=0}^{k-1} (\phi_1^2)^j \quad \leftarrow \text{sum of a GP. } k \text{ terms.}$$

$$= \sigma_z^2 \left(\frac{1 - \phi_1^{2k}}{1 - \phi_1^2} \right).$$

$$4(a) \quad X_t = \sum_{m=0}^{\infty} b_m Z_{t-m}$$

$$= \sum_{m=0}^{\infty} b_m B^m Z_t \quad (*)$$

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j}$$

$$= \sum_{j=0}^{\infty} d_j(k) B^j X_n$$

$$= \sum_{j=0}^{\infty} d_j(k) B^j \left(\sum_{m=0}^{\infty} b_m B^m Z_n \right)$$

「Because of (*)」

$$= \left[\sum_{j=0}^{\infty} d_j(k) B^j \right] \left[\sum_{m=0}^{\infty} b_m B^m \right] Z_n$$

$$= B(k\xi)$$

$$= D_k(\xi) B(\xi) Z_n \quad \text{for } \xi = B \quad (1)$$

$\uparrow$
 B/ward
 shift
 op

$\hat{X}_{n+k}(n)$ needs to be such that

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} b_{j+k} Z_{n-j}$$

$$= \sum_{j=0}^{\infty} b_{j+k} B^j Z_n$$

$$= \left(\sum_{j=0}^{\infty} b_{j+k} B^j \right) Z_n$$

$$= C_k(\xi) Z_n \quad \text{for } \xi = B. \quad (2)$$

Equating (1) and (2), we see that

$$C_k(\xi) = D_k(\xi) B(\xi).$$

(b) It is of importance because the optimal predictor can then be expressed in terms of present/past values of the series rather than white noise terms.

$$6. \quad \hat{X}_{t+1} = \sum_{r=0}^{\infty} a_r X_{t-r}$$

Consider

$$E[(X_{t+1} - \hat{X}_{t+1})^2]$$

$$\begin{aligned} (X_{t+1} - \hat{X}_{t+1})^2 &= \left(X_{t+1} - \sum_{r=0}^{\infty} a_r X_{t-r} \right)^2 \\ &= \left(Z_{t+1} + \theta Z_t - \sum_{r=0}^{\infty} a_r (Z_{t-r} + \theta Z_{t-r-1}) \right)^2 \\ &= \left(Z_{t+1} + (\theta - a_0) Z_t \right. \\ &\quad \left. - (a_1 + \theta a_0) Z_{t-1} \right. \\ &\quad \left. - (a_2 + \theta a_1) Z_{t-2} \right. \\ &\quad \left. - \dots \right)^2 \\ &= \left(Z_{t+1} + (\theta - a_0) Z_t \right. \\ &\quad \left. - \sum_{r=1}^{\infty} (a_r + \theta a_{r-1}) Z_{t-r} \right)^2 \end{aligned}$$

The above term is a square and ~~with~~ its expectation will be minimised where

$$\theta - a_0 = 0 \quad \Rightarrow \quad a_0 = \theta$$

$$a_r + \theta a_{r-1} = 0 \quad \Rightarrow \quad a_r = -\theta a_{r-1} \\ \text{for } r = 1, 2, 3, \dots$$

$$a_0 = \theta$$

$$a_1 = -\theta a_0 = -\theta^2$$

$$a_2 = -\theta a_1 = \theta^3$$

$$a_3 = -\theta a_2 = -\theta^4$$

$$\text{Generally } a_r = (-1)^r \theta^{r+1}$$

$$\therefore \hat{X}_{t+1} = \sum_{r=0}^{\infty} (-1)^r \theta^{r+1} X_{t-r}$$

3. Consider $Y_t = \nabla X_t$
 $= (1-B)X_t$

We have $X_t \leadsto (1-\phi B)Y_t = Z_t$

$\Rightarrow \{Y_t\}$ is an AR(1) process.

$\{Y_t\}$ is (weakly) stationary.

Set $(1-\phi B)\hat{Y}_{n+k}(n) = [Z_{n+k}]$

where $[Z_k] = \begin{cases} 0 & \text{if } k > n \\ Z_k & \text{otherwise, } k \leq n. \end{cases}$

$k=1$:

$$(1-\phi B)\hat{Y}_{n+1} = 0$$

$$\Rightarrow \hat{Y}_{n+1} = \phi Y_n$$

$k=2$:

$$(1-\phi B)\hat{Y}_{n+2} = 0$$

$$\hat{Y}_{n+2} = \phi \hat{Y}_{n+1}$$

$$= \phi^2 Y_n$$

Generally $\hat{Y}_{n+k} = \phi^k Y_n.$

$$\hat{X}_{n+k} :$$

Consider

$$\hat{X}_{n+1} = X_n + \hat{Y}_{n+1}$$

$$\begin{aligned}\hat{X}_{n+2} &= \hat{X}_{n+1} + \hat{Y}_{n+2} \\ &= X_n + \hat{Y}_{n+1} + \hat{Y}_{n+2}\end{aligned}$$

Generally

$$\hat{X}_{n+k} = X_n + \sum_{r=1}^k \hat{Y}_{n+r}(n)$$

$$= X_n + \sum_{r=1}^k \phi^r Y_n$$

$$= X_n + \frac{\phi(1-\phi^k)}{1-\phi} Y_n$$

$$= X_n + \frac{\phi(1-\phi^k)}{1-\phi} (X_n - \cancel{X_{n-1}}^{\cancel{X_{n-1}}})$$